Chapter 10 — Electrochemistry

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Redox reactions and oxidation numbers

Q.1 Explain oxidation, reduction, disproportionation and oxidation number rules.

Oxidation is loss of electrons or increase in oxidation number; reduction is gain of electrons or decrease in oxidation number. A redox reaction involves both processes. Disproportionation is a reaction in which the same species is both oxidized and reduced, such as Cl_2 in cold NaOH forming Cl^- and ClO^-.

Oxidation number rules include: free elements are zero; monatomic ions equal their charge; oxygen is usually –2; hydrogen is usually +1; fluorine is –1; the sum equals the overall charge. The oxidizing agent is reduced, while the reducing agent is oxidized.

Important Conceptual Short Questions — Topic 1

1. Why is oxidation always accompanied by reduction?

Electrons lost by one species must be gained by another. Therefore oxidation and reduction occur together in a redox reaction.

2. How does oxidation number help identify the oxidizing agent?

The oxidizing agent contains an element whose oxidation number decreases, meaning it gains electrons and is reduced.

3. Why is chlorine disproportionation in cold alkali a redox reaction?

Chlorine starts at oxidation state 0 and forms both Cl^- at –1 and ClO^- at +1. The same element is both reduced and oxidized.

4. How can oxidation number rules be used to balance a redox equation?

Identify the elements changing oxidation number, make the total increase equal to the total decrease, then balance atoms and charge.

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Topic 2 — Electrolytic cells and electrolysis

Q.2 Describe electrolytic cells and compare electrolysis of molten and aqueous electrolytes.

An electrolytic cell converts electrical energy into chemical energy. Oxidation occurs at the anode and reduction at the cathode. In molten NaCl, Na^+ is reduced to Na at the cathode and Cl^- is oxidized to Cl_2 at the anode.

In aqueous solutions, water may also be oxidized or reduced. The product depends on electrode potentials, concentration and nature of electrodes. In concentrated NaCl solution, chlorine is produced at the anode, hydrogen at the cathode and NaOH remains in solution.

Important Conceptual Short Questions — Topic 2

1. Why does molten NaCl give sodium metal but aqueous NaCl gives hydrogen?

In molten NaCl, only Na^+ and Cl^- are present. In aqueous NaCl, water is easier to reduce than Na^+, so hydrogen forms at the cathode.

2. How does the nature of electrode affect electrolysis products?

An inert electrode does not take part, but an active electrode may dissolve or react, changing the products.

3. Why is oxidation always at the anode?

By definition, the anode is the electrode where electrons are released by oxidation, whether the cell is electrolytic or galvanic.

4. How does concentration affect discharge of ions during electrolysis?

A high concentration of an ion can favour its discharge even if another ion is slightly easier to discharge under standard conditions.

Topic 3 — Quantitative electrolysis and Faraday's laws

Q.3 Explain the quantitative relationships in electrolysis and how Avogadro's constant can be found by electrolysis.

The charge passed is Q = It, where I is current in amperes and t is time in seconds. One mole of electrons carries one faraday, F ≈ 96500 C mol^{-1}. The amount of substance deposited is proportional to the charge passed and inversely proportional to the number of electrons in the electrode reaction.

For a metal ion M^{n+}, moles of metal = Q/(nF). Avogadro's constant can be estimated by measuring the mass deposited by a known charge and using the charge on one electron: $N_A = F/e$.

Important Conceptual Short Questions — Topic 3

1. Why does the same charge deposit different masses of different metals?

Different metals have different molar masses and require different numbers of electrons per ion. Therefore the mass deposited per faraday differs.

2. How is the number of faradays related to the electrode equation?

The coefficient of electrons in the half-equation gives the faradays needed per mole of product. For $Cu^{2+} + 2e^- \rightarrow Cu$, two faradays deposit one mole of copper.

3. Why is current efficiency important in electrolysis?

If some current produces side reactions, the actual mass deposited is less than the theoretical mass calculated from Q = It.

4. How can electrolysis be used to estimate Avogadro's constant?

Measure the charge needed to deposit a known amount of substance, calculate the charge per mole of electrons, and divide by the charge of one electron.

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Topic 4 — Electrode potentials and standard hydrogen electrode

Q.4 Explain the standard hydrogen electrode and measurement of standard electrode potentials.

When a metal is placed in a solution of its ions, an electrical double layer forms and a potential difference develops. The standard hydrogen electrode consists of platinum black in contact with H_2 gas at 1 atm and H^+ ions at 1 mol dm^{-3}, at 298 K. Its standard electrode potential is assigned 0.00 V.

When another half-cell is connected to SHE, the measured cell voltage gives its standard electrode potential. A positive E° means the electrode is reduced more readily than H^+; a negative E° means it is oxidized more readily. Copper has E° = +0.34 V and zinc has E° = –0.76 V.

Textbook check (p. 230): the worked example prints E° for Ag^+/Ag as +0.080 V in one place; the standard value used elsewhere is +0.80 V. Also, some reduction half-equations are labelled E°(ox); use reduction potentials consistently.

Important Conceptual Short Questions — Topic 4

1. Why is platinum used in the standard hydrogen electrode?

Platinum is inert, conducts electricity and is coated with platinum black to adsorb hydrogen gas and catalyse the electrode reaction.

2. How does the sign of E° indicate tendency for reduction?

A more positive E° means a greater tendency to gain electrons and be reduced. A more negative E° means a greater tendency to lose electrons and be oxidized.

3. Why is SHE assigned zero potential?

A single electrode potential cannot be measured alone. SHE is chosen as a universal reference and assigned 0.00 V so other potentials can be measured relative to it.

4. How does an electrical double layer form?

Metal atoms may leave the surface as ions, leaving electrons on the metal, or ions may deposit. This charge separation attracts oppositely charged ions and forms a double layer.

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Topic 5 — Galvanic cells, salt bridge and applications of E°

Q.5 Describe a galvanic cell and explain applications of standard electrode potentials.

A galvanic cell converts chemical energy into electrical energy. In a Cu-Zn cell, zinc is oxidized at the anode: $Zn \rightarrow Zn^{2+} + 2e^-$; copper ions are reduced at the cathode: $Cu^{2+} + 2e^- \rightarrow Cu$. Electrons flow from zinc to copper through the wire.

The salt bridge completes the circuit and maintains electrical neutrality by allowing ion flow. E° values predict direction of electron flow, feasibility of redox reactions and oxidizing/reducing strength. A positive cell potential indicates a

feasible reaction under standard conditions.

Important Conceptual Short Questions — Topic 5

1. Why does the voltage of a galvanic cell fall to zero without a salt bridge?

Charge builds up in the half-cells and stops further electron flow. The salt bridge prevents this by allowing ions to migrate and maintain neutrality.

2. How can E° values predict the direction of electron flow?

Electrons flow from the half-cell with the more negative reduction potential, which acts as the anode, to the half-cell with the more positive reduction potential, the cathode.

3. Why does zinc act as the anode in a Cu-Zn cell?

Zinc has a more negative reduction potential than copper, so it is more easily oxidized and releases electrons.

4. How does a salt bridge maintain electrical neutrality?

Anions move toward the anode compartment to balance excess positive charge, and cations move toward the cathode compartment to replace reduced cations.

Topic 6 — Nernst equation, activity series and photovoltaic cells

Q.6 Explain the Nernst equation, activity series and photovoltaic cells.

The Nernst equation relates electrode potential to ion concentration. At 298 K for a reduction half-reaction, $E = E° + (0.059/n) \log[\text{ion}]$. It shows that potential changes with concentration. The activity series arranges metals by their tendency to lose electrons; a more reactive metal can displace a less reactive metal from its salt solution.

Photovoltaic cells convert light energy directly into electrical energy using semiconductor materials. Photons excite electrons across a band gap, and a p-n junction separates charges. They are a sustainable energy source because they use sunlight and produce no emissions during operation.

Important Conceptual Short Questions — Topic 6

1. Why does electrode potential change with ion concentration?

Changing ion concentration changes the tendency of the electrode reaction to proceed. The Nernst equation quantifies this shift from the standard potential.

2. How can the activity series predict displacement reactions?

A metal higher in the series can lose electrons more easily and displace a lower metal from its salt solution.

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3. Why is a photovoltaic cell different from a galvanic cell?

A photovoltaic cell converts light energy directly into electricity using a semiconductor, while a galvanic cell converts chemical energy from a spontaneous redox reaction.

4. How does a p-n junction help a photovoltaic cell produce current?

The internal electric field at the junction separates photo-generated electrons and holes, driving them in opposite directions and producing a voltage.

Topic 7 — Winkler method, BOD and DO

Q.7 Explain dissolved oxygen, biochemical oxygen demand and the Winkler method.

Dissolved oxygen (DO) is the amount of oxygen dissolved in water and is essential for aquatic life. Biochemical oxygen demand (BOD) is the amount of oxygen required by microorganisms to decompose organic matter in water, usually over five days. High BOD indicates organic pollution and can lower DO.

The Winkler method determines dissolved oxygen. Oxygen fixes manganese(II) in alkaline solution, and acidification liberates iodine equivalent to the oxygen present. The iodine is titrated with sodium thiosulfate using starch indicator.

Important Conceptual Short Questions — Topic 7

1. Why does high BOD indicate polluted water?

A high BOD means there is much biodegradable organic matter. Microorganisms consume dissolved oxygen while decomposing it, which can suffocate aquatic life.

2. How can low dissolved oxygen harm aquatic life?

Fish and other aerobic organisms need dissolved oxygen for respiration. Low DO causes stress, reduced growth and death.

3. Why is the Winkler method used in water analysis?

It gives a reliable quantitative measure of dissolved oxygen through a redox titration, allowing water quality to be assessed.

4. How does organic waste increase BOD?

Organic waste provides food for microorganisms. Their respiration consumes oxygen, increasing the biochemical oxygen demand.

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